Ikigai -- A Biologically Grounded Digital Organism: Development Report, Day 80

Mura ALife Labs (Formerly Hitoshi AI Labs) -- NeuroSeed Project

Author: Prince Siddhpara, Founder -- Mura ALife Labs (Formerly Hitoshi AI Labs) **Date:** June 24, 2026 **Project:** NeuroSeed **Subject:** Day 80 -- The Framework Day. 15 Packs Shipped (307-321), Zero Reverted, 30/30 Held. THE NORTH-STAR SHIFT (Build The FRAMEWORK, Not A Data-Loaded Organism). Pack 307 Multi-Word Answers + Pack 308 Groq Instruct Teacher (SimpleQA 0%->7%, 3.5x). Pack 309 + 310 CALIBRATION (Teacher-Side Abstention + The Geometry-Derived Boundary k/sqrt(2d) -- Framework Line #11, The LLM Differentiator, Drawn). FREE FLUENCY SOLVED DATA-FREE (Packs 311-316): Product-of-Experts AND-Gate Breaks The Token Attractor; Frame-Then-Fill Parallel Relaxation Beats Next-Token Prediction; Distributional POS Induction With No Labels; WHOLE-TEMPLATE Frames BREAK The Markov Frame-Mixing Wall -> 100% Grammatical English; Wired As An Organism API (fit_free_fluency + say_frame). Pack 317 + 317.2 DERIVE-NOT-STORE Taken To Its Limit (Arbitrary N-Hop, Wildcard Inheritance Schema, Transitivity Discovery, Autonomous Rule-Mining In Sleep). Pack 318 Self-Termination + Unbounded Multi-Hop (The Organism Stops Itself; Hops As Deep As The Data). Pack 319 LIMITS (Operating Envelope: ~20k Facts/Role, Composition Unbounded At 100 Hops, A Real Calibration Flaw Caught). Pack 320 EMPIRICAL CALIBRATION (Honest-Unknown Held Under Load). Pack 321 THE COMPRESSION MULTIPLIER MEASURED (1 Billion Compositional Facts <- ~104k Atoms ~= 570 KB). Framework Audit: 11/11 Mechanisms Drawn. Two Peer Reviews (CCM, Proto). The EBM Reframe. Zero Hardcoding (Enforced Again). **Classification:** Research Document -- Computational Neuroscience / Artificial Life / Cognitive Systems Engineering

---

1. Objective

Day 79 closed at fifteen packs and the compositional turn: store atoms, derive relations, discover the rules, speak the facts. Day 80 opened on the locked queue -- multi-word answer storage and a better teacher to lift the honest 2% SimpleQA baseline -- but its arc was set by a mandate Prince delivered mid-day that reframed everything that followed.

The mandate: **build the FRAMEWORK, not a data-loaded organism.** Prove every load-bearing line of the architecture data-free, with made-up entities, so that what remains when you remove the data is a complete generative brain rather than a memorised lookup table. "Framework without data. If you feed it the data it can do it -- but without data it is just a framework." The differentiator versus large language models is not knowledge volume; it is calibration, composition, honesty, and footprint. The day's work reverse-engineered from an eleven-line map of the framework, and by close all eleven lines were drawn.

Two questions from Prince shaped the late day and are recorded here as first- class results: how meaning binds in the substrate (answer: the shared computed key -- one token is one phasor, a hub for every relation bound to it), and whether a billion facts can live in 500 MB (answer: not as random facts -- Shannon forbids it -- but as compositional facts, yes, by storing the square- root kernel and deriving the rest, a multiplier we then measured).

Day 80 ended at fifteen packs shipped, none reverted, the substrate-only bench holding **30/30 = 100%** throughout. Every pack was code-only; no `organism.ikg` save was taken, in keeping with the framework-first doctrine.

---

2. Pack 307 -- Multi-Word Answer Storage

The Day 79 SimpleQA bottleneck was diagnosed not in storage but in the PARSE. The anchor cache already stored the action token tuple whole and joined it on recall ("buenos aires" already worked); the gap was `TeacherOracle._parse_answer` collapsing every answer to its single last content token ("south america" -> "america", "may 16 2015" -> "2015"). A `multiword=True` flag now returns the full content phrase (same echo/stopword drop, no last-word collapse) with a matched system prefix, wired through `integrate.enable_active_learning` (max_new_tokens 12 -> 32 so the phrase is not truncated). Single-word stays the default, so 30/30 and every existing caller are untouched. Gate: multi-word parse correct; an end-to-end made-up "zogland" active-learning run cached and recalled whole phrases substrate-only; capitals 13/13, math 9/9. One honest limit: a lone initial "a" is dropped by the legacy article stopword set (distinguishing article-a from initial-a would need a heuristic = hardcoding), left as-is.

---

3. Pack 308 -- Groq Instruct Teacher + SimpleQA Re-Run

The teacher was swapped from the 3090 vLLM R1-distill to the Groq cloud API. A new `GroqTeacher` mirrors the subset `TeacherOracle` drives and fans batch requests over a thread pool (Groq has no batch endpoint). The reason was qualitative: Groq serves INSTRUCT models that return clean answers ("buenos aires", "south america", "pound sterling") with no `<think>` ramble, killing the whole class of parse disease that plagued the R1 distill. Locked to 8B (`llama-3.1-8b-instant`) at Prince's explicit instruction. Result: real-SimpleQA 0% cold -> **7%** post-teach with multi-word storage, against the Day 79 baseline of 0% -> 2% -- a 3.5x lift. The 7% is the honest 8B ceiling on obscure long-tail facts; failures split into teacher-wrong and teacher-abstains, the latter motivating the next two packs.

---

4. Pack 309 -- Abstention Gate (Teacher Side)

When the 8B teacher does not know, it should not be allowed to pour a guess into the cache. The gate is a PROTOCOL sentinel, not a phrase list: the prompt instructs "if you do not know, reply with exactly `idk`", and `ask_batch` drops any answer equal to the sentinel so it is never cached. This is instruction- following (like "answer in one word"), not a hardcoded blacklist, and it dodges the Day 79 self-confirmation sycophancy trap because it detects abstention rather than verifying answers. Re-running 303 with the gate moved the headline 7% -> 3% -- not a regression but an honesty correction: the old 7% was inflated by lucky token overlaps ("october 5 2015" scoring on "2015"); removing guesses lowers the number and raises its truth.

---

5. Pack 310 -- Calibration: The Boundary (Framework Line #11)

The differentiator versus LLMs, drawn as a data-free mathematical line. An LLM hallucinates confidently; a 200 MB brain that returns "unknown" and is RIGHT is more trustworthy than a 400 GB model that bluffs. The boundary is pure geometry: in FHRR a cleanup similarity for two independent (absent-fact) HVs has mean 0 and standard deviation $1/\sqrt{2d}$, so the abstain boundary is $k/\sqrt{2d}$, with k a confidence multiple, not a tuned magic number. At $d=400$, $k=4$ gives boundary 0.1414 and a per-candidate false-accept of 3×10^{-5}. A new `calibration.py` provides the formulas; `general_reasoner.reason(do_abstain=True)` replaces the old hand-tuned `icl_min_state_sim=0.10` with the derived value and returns "unknown" when recall sits at the noise floor, before the next-token guess that would otherwise be the hallucination. Notably, a first attempt gated on an interrogative word list; Prince flagged it as hardcoding and it was reverted -- the gate is geometry confidence, not word shape. Gate: four made-up entities abstain; with abstention off, four hallucinate a noise token (proving the line changes behaviour); capitals 13/13, math 9/9.

---

6. The North-Star Shift -- Framework Over Data

Mid-day Prince reframed the whole project. Stop accumulating features; draw the mathematical lines of the framework data-free. An eleven-line map was written into memory as the reverse-engineering target, with calibration identified as the missing eleventh line and the true differentiator. The emotional stakes were stated plainly and recorded: a seventeen-year-old solo researcher, eight years of work, the goal of a phone-local brain as capable as the best frontier model in 500 MB. Everything from here was built to that map. The remaining sections are the lines being drawn.

---

7. Free Fluency I -- The Product-of-Experts AND-Gate (Packs 311)

The Day 79 free-fluency negative was the function-word attractor: high-frequency words dominate the superposed transition bank, so a greedy argmax walk collapses into "the of the of." A standalone FHRR probe isolated the fix from an external research report. PMI reweighting alone does nothing (the dominant token's mass survives the divide), but a holographic PRODUCT-OF-EXPERTS -- multiplying the next/next2/next3 similarity profiles instead of summing them -- is the load- bearing mechanism: a token survives the fused score only if it is supported across ALL orders, so longer context overrides the unigram prior. Wired into the real bank as `MultiRoleMemory.poe_candidates` and `IkigaiOrganism.say_free`. On the real 182 MB organism the collapse reproduced and PoE+PMI won decisively (in-sentence repeat 69% -> 38%, "to find the number of" emerging), with PMI load-bearing here because the real Zipfian skew is extreme. The single-token attractor was broken; phrase-level looping remained, pointing to the next layer.

---

8. Free Fluency II -- Frame-Then-Fill Parallel Relaxation (Packs 312, 313)

The deeper fix, and the substrate form of "do what a real brain does." Not autoregression: fix a MESSAGE, lay a syntactic FRAME (a category sequence), then FILL every slot at once by bidirectional parallel relaxation, each slot scored from both neighbours under a category mask, iterated to a fixpoint. Function words come from the frame's STRUCTURAL slots, not from frequency prediction, so they cannot form an attractor -- this is Levelt's message-first model in vector algebra. The concept probe took grammaticality 0% -> 100% and repeat 50% -> 0% on a made-up grammar. Pack 313 wired it as `frame_relax.py` and `IkigaiOrganism.say_frame`, running the relaxation over the real recall API with backward evidence approximated by reverse-querying the forward bank (production has no backward bank). A key finding on the real SDM bank: greedy does not repeat- loop there (the SDM prevents it) -- it MODE-COLLAPSES, emitting the same sentence every time; frame-relaxation restored diversity (unique 27% -> 100%) at the corpus function-word baseline.

---

9. Free Fluency III -- Distributional POS Induction (Pack 314)

Frame-then-fill needs SYNTACTIC categories, and the organism only had semantic ones (the FrameField clusters words by domain, not part of speech). The fix is classical and substrate-native: cluster words by their left-and-right neighbour distributions, the Harris/Brown distributional hypothesis, with no labels. The probe recovered the toy grammar's parts of speech at 100% purity from counts and 90% through FHRR recall. Wired on the real organism, induction WORKED -- a clean function-word cluster (the/is/to/of/and/a), a pronoun cluster (i/we/you) -- genuine syntactic separation the domain field never had. But generation degenerated to "the the of of": the recall-derived category FSM was self-loop heavy and the fill repeated within a category. The induction was banked; the category-level attractor was the next wall, and it was diagnosed honestly rather than band-aided.

---

10. Free Fluency IV -- Breaking The Markov Frame-Mixing Wall (Packs 315, 316)

Pack 315 added an anti-repeat fill and a clean-data pipeline, lifting readable English but capping grammaticality at 17% -- a finite-order category Markov FSM MIXES frames, because the state (DET,NOUN) is both a sentence start and a sentence end, so the walk jumps start-to-end or loops. Prince's instinct was that this is not a fluency ceiling but a wrong model, and he was right. Pack 316 bypassed it: store WHOLE observed category sequences as atomic TEMPLATES (what the positional roles are for) and retrieve a complete frame, then fill it. A whole template can never stitch a start to an end, so mixing is structurally impossible. Result on clean data: grammaticality 17% -> **100%**, real English -- "the cold child chases", "the old dog likes a dog", "a fish likes a small fish". The wall was the frame model, not fluency. Wired end-to-end as `fit_free_fluency(texts)` + `say_frame`, 92% grammatical on a trained organism. Free-fluency grammar is solved data-free; the shipped organism's fluency is now gated only on clean prose data, which is a data axis, not a mechanism gap.

---

11. Pack 317 -- Derive-Not-Store Taken To Its Limit

Two upgrades to the composition engine, both derive-read-only. Arbitrary N-HOP chaining generalises the Day 79 two-hop to any depth: `_parse_chain` strips "the R of" prefixes and `_chain_resolve` resolves innermost-out, each hop a direct atom or a learned inverse. WILDCARD INHERITANCE anti-unifies two or more per-attribute inheritance rules into a single schema, attr='*', meaning "a capital inherits ALL its country's attributes" -- the organism generalises "this attribute inherits" to "all attributes inherit", a far stronger move than fuzzy cosine matching could give (different attribute tokens have different HVs). Gate: 3-hop made-up chains resolve; with require_learned on, an untaught continent-of-capital is unknown until the wildcard is promoted, then it derives. Regression 304.2 9/9, capitals 13/13, math 9/9.

---

12. Pack 317.2 -- Transitivity Discovery + Autonomous Sleep Mining

`RuleMiner.mine_transitive` discovers that a relation is transitive from atoms alone -- R(a)=b and R(b)=c chains, acyclic, above a support threshold -- and promotes a transitive rule. The engine's `transitive_reach` then follows the relation to its root, computing the closure ON DEMAND rather than storing every ancestor pair: derive-not-store at the relation level. The autonomous piece: `sleep_consolidate` now calls `discover_rules` every sleep, so the organism mines inheritance, synonymy, inverse, and transitive rules while resting. The end-to- end gate is the signature result: train a flat organism on a made-up chain, run one sleep cycle, and the organism discovers the relation is transitive (confidence 1.0) entirely on its own, then derives the four-hop closure -- no manual mining call.

---

13. Pack 318 -- Self-Termination + Unbounded Multi-Hop

Two of Prince's "infinite" wishes, drawn. The organism stops ITSELF: `say_free` emits END on two self-generated signals -- the calibration boundary (no confident continuation) and cycle detection (a repeated bigram means no new content). A genuine finding emerged here -- on the SDM transition bank an unwritten successor does not read as noise but as crosstalk-fabricated confidence, so the boundary alone does not transfer to transitions (it DOES work for fact recall, the actual "capital of France -> Paris, stop" path); cycle detection is the robust transition stop. And multi-hop is now CONVERGENCE-bounded, not fixed-cap: `transitive_reach` follows the chain until the data runs out, resolving a 14-hop made-up chain and stopping at its root. The organism decides when it is done and hops as deep as the structure goes -- we never impose a count.

---

14. Pack 319 -- The Limits: Operating Envelope

With data procurement weeks away, the framework was stress-tested to find its cliffs first. Three probes. CAPACITY: recall holds 100% to ~20k facts per role at $d=400$ and degrades to 77% at 50k -- a location-crowding wall of the current SDM configuration, not the dimensional limit, liftable by scaling d or sharding. COMPOSITION DEPTH: unbounded -- 100-hop chains resolve perfectly, because derive is exact symbolic over the atom index, with no per-hop crosstalk. CALIBRATION UNDER LOAD: a real flaw caught -- the boundary $k/\sqrt{2d}$ is the per-comparison floor, but cleanup takes the argmax over N candidates, and the maximum of N noise draws exceeds it, giving 98% false-accept at N=1000. A multiple-comparison correction (`abstain_boundary_n` $= \sigma\sqrt{2\ln(N/\alpha)}$) drove false-accept to zero at scale but left a residual at small N -- the SDM crosstalk again.

---

15. Pack 320 -- Empirical Calibration: Honest-Unknown Under Load

The crosstalk residual is closed substrate-natively. `MultiRoleMemory.calibrate_abstain` probes guaranteed-absent random keys against the bank, measures their cleanup-similarity distribution, and sets the boundary to mean + $k\cdot$std -- the same k-sigma form, but with σ MEASURED from the bank's own noise (crosstalk included) rather than derived from d. Nothing is tuned; the threshold is read off the substrate. Result: false-accept ~0% at every load (1k, 5k, 20k), where the theoretical floor leaked 98% and even the N-correction leaked 21% at small N, with false-reject 0%, and the boundary adapts (0.255 where crosstalk is high, 0.19 at scale). Honest-unknown -- the LLM differentiator -- now holds under load.

---

16. Pack 321 -- The Compression Multiplier, Measured

The direct answer to Prince's billion-facts-in-500-MB question. Effective capacity = stored atoms $\times$ derivation fanout. A made-up world of countries with capitals, continents, currencies, and languages was stored as atoms plus one inverse rule; every derivable fact -- inheritance, inverse chains, and the quadratic space of same-attribute comparisons -- was probed and VERIFIED against ground truth. Stored-to-derived: 41 -> 165 (4.0x), 81 -> 630 (7.8x), 161 -> 2460 (15.3x). The ratio is proportional to the knowledge size, because the derivable comparison space is $O(C^2)$ while the stored kernel is $O(C)$. Inverting: N derivable facts need ~$3.3\sqrt{N}$ stored atoms, so **one billion compositional facts follow from ~104k atoms, about 570 KB.** A billion facts under a megabyte -- not by beating Shannon (random facts remain forbidden), but by storing the square-root KERNEL and deriving the rest, with the multiplier GROWING at scale. The honest condition is stated plainly: this holds for compositional facts, and real-world knowledge is overwhelmingly compositional.

---

17. Two Peer Reviews and the Energy-Based Reframe

Two external papers were reviewed at Prince's request. The Cognitive Coherence Model (CCM, a society member's architecture) was assessed as one strong idea -- misalignment between somatic and mental engines as a primary datum -- wrapped in theatrical geometry and a hand-authored knowledge graph, the exact failure modes NeuroSeed engineers against; a respectful reply was drafted. Proto (Stanford/Arc, a high-level language for generative biology) was the consequential one: its core abstraction is an energy-based model, $\pi(x)\propto p(x)\exp(-f(x)/T)$, with multi-objective design as a PRODUCT OF EXPERTS -- exactly the Pack 311 AND-gate and the Pack 313 relaxation, independently arrived at and validated at field scale with wet-lab results. This produced a reframe, saved to memory: NeuroSeed is an energy-based design language over a holographic substrate -- generator (recall), constraints (calibration, grammar, PMI, arithmetic), optimizer (relaxation/annealing) -- unifying the eleven lines under one primitive set.

---

18. Meaning, Self-Termination, and the Framework Audit

Prince's deep questions were investigated against the files, not hand-waved. MEANING binds via the shared computed key: `key('paris')` is generated from the string, identical everywhere, so one token is one phasor HV and a hub for every relation bound to it; meaning is the superposition of all those bindings, and richness per token is simply how many relations are bound -- a data question, not a mechanism gap. The eleven-line map was audited: FHRR, SDM body, the RHC number ring, cat-1-4 ingest, composition, logic closure, learning, compression, language (mechanism complete), consolidation, and calibration -- **11/11 mechanisms drawn**, with only line nine's prose DATA deferred, which Prince explicitly scoped out (100% grammar is data, not structure).

---

19. Close -- Day 80 Metrics

Fifteen packs shipped (307 through 321), none reverted, the substrate-only bench holding **30/30 = 100%** through every landing. All code-only; no `organism.ikg` save was taken, per the framework-first doctrine.

The day's signature was completion: an eleven-line framework map drawn end to end, data-free, with made-up entities throughout. Calibration (line eleven) was drawn and then HARDENED under load. Free-fluency grammar was solved -- the function-word attractor broken at the token level, the phrase level, and finally the frame level, ending in 100% grammatical English from whole-template frames and a working organism API. Derive-not-store was taken to its limit -- arbitrary hops, a wildcard inheritance schema, transitivity discovered autonomously in sleep -- and the compression bet underneath it all was measured: a billion compositional facts from a square-root kernel under a megabyte.

The framework is structurally complete. What remains is the data axis Prince named: feed a one-to-two-million-fact structured corpus -- Wikidata, ConceptNet, the world's existing knowledge graphs, not hand-fed -- let the organism discover the rules in sleep and self-compress to the kernel, and watch the kid talk and reason. The hard part, the framework, is built.

Production state: organism.ikg 182 MB (untouched -- no save this day); cache and learned rules as Day 79 (code-only day); substrate-only bench 30/30 = 100% held throughout; zero hardcoding on the production path, enforced again. New modules: `calibration.py`, `frame_relax.py`. New persisted memory: the framework map, the operating envelope, the EBM reframe, the free-fluency research record. One standing outbound obligation: the Kanerva bump on 2026-07-01 if no reply.

Trigger keyword for post-compact memory reload: `DAY 80 FRAMEWORK COMPLETE`.